I Tutorial

Prices, return distributions and rolling inference

Exercise 1. Prices, returns and empirical distributions

Notebook: Prices_Returns_Distributions.ipynb

Analyse daily equity, energy and exchange-rate data (2010–2025). Compare levels and simple/log returns, identify volatility clustering, and compare return distributions across markets and with the Normal benchmark. Comment on expectations and observed evidence.

Relevant concepts

- Prices versus returns.
- Returns (simple, log and cumulative); total and excess returns; time aggregation.
- Volatility and volatility clustering.
- Characteristics of financial return distributions (skewness, excess kurtosis and fat tails).
- Comparison with the Normal distribution; standardisation; histograms and Q–Q plots.

To do. Choose an asset with a regularly observed price series (e.g., a stock, an ETF or a cryptocurrency). Repeat the notebook's analysis of prices and returns, and compare the return distribution with the Normal benchmark.

Exercise 2. Rolling statistics and the equity risk premium

Notebook: Rolling_Equity_Premium.ipynb

Analyse monthly US market excess returns for 2011–2025. Compare rolling means and standard deviations using 30-, 40-, 50- and 60-month windows. Use the 60-month two-sided t-test to assess evidence against a zero mean excess return, and discuss the economic interpretation and limitations of the results.

Relevant concepts

- Equity risk premium; expected excess return versus sample average.
- Rolling mean and volatility; effect of window length.
- Hypothesis testing (null and alternative hypotheses, standard error and t-test).
- P-values; significance levels; Type I error; rejection and non-rejection.
- Statistical versus economic significance.
- Test assumptions and limitations; overlapping windows and repeated testing.

To do. For the final month in the notebook's 60-month results table, report the mean excess return and p-value. State H_0 and H_1, interpret the test at 5%, and distinguish statistical significance from the size of the estimated premium.

Exercise 3. Tail risk and Value at Risk (optional)

Notebook: Tail_Risk_VaR.ipynb

Compare historical and fitted Normal 1% lower-tail thresholds for the US market, Brent and EUR/USD, and relate them to 99% one-day VaR. Examine crossings in 2020–2025 using thresholds estimated in 2010–2019 and held fixed.